

Tour — Akgül et al. 2026: Sparse Policy Selection for LLM Reasoning

Learning-map artifact

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Abstract

A guided learning path through Akgül et al. 2026, the foundation concepts it depends on, and the constructive extensions worth building. Companion to `tour.md` and `tour.ipynb`.

Section 1 — Reader’s Contract

Who this is for. Mathematicians, ML researchers in adjacent subfields (optimisation, probability, statistical learning), and engineers comfortable reading per-token training equations. The paper is empirical-mechanistic with a thin but sharp probabilistic spine; you will get more out of it if you are comfortable manipulating expectations over discrete distributions and reading KL terms without flinching.

Background assumed. Linear algebra at the level of eigenvalues and linear maps; multivariate calculus through the Jacobian; probability through random variables, expectation, and variance; Shannon information at the level of entropy and KL. No measure theory required for the body.

Four pre-built foundation tours are available; we recommend the math-grad tour as the entry point for this paper:

Audience	Tour
Novice	https://github.com/pleyva2004/math-foundations/blob/main/tours/nov
CS undergrad	https://github.com/pleyva2004/math-foundations/blob/main/tours/cs-
Math grad (recommended)	https://github.com/pleyva2004/math-foundations/blob/main/tours/math-
Researcher	https://github.com/pleyva2004/math-foundations/blob/main/tours/res

Expected reading time. 3–4 hours for a focused first pass.

One-sentence elevator. Akgül et al. show that RL post-training on verifiable rewards (RLVR) for math reasoning is *not* learning new capabilities — it is performing a sparse, entropy-gated re-selection at roughly the top 2–8% of token positions, where the RL policy promotes a token that the base model already had at mean rank ≈ 2.2 , and the entire effect can be reproduced by an offline contrastive fine-tune (*ReasonMaxxer*) with no rollouts.

Section 2 — Foundations Walk

Topological order: probability before information theory before linear algebra applied to gradients. Levels come from the math-foundations manifest. “Why this paper needs it” is sourced from `02-math-deep-dive.md`.

#	Concept	Level	Pacing	Why this paper needs it
06	Linear Maps and Matrices	intermediate	skim	Attention projections W_Q, W_K, W_V, W_O are linear maps; LoRA delta $\Delta W = BA$ is a low-rank decomposition.
07	Eigenvalues and Eigenvectors	intermediate	skim	Spectral structure of the rank-32 LoRA delta and the operator-norm step in the proposed Lipschitz bound.
14	Gradient and Jacobian	advanced	read	Contrastive loss and base-anchoring KL are differentiated through the softmax Jacobian.
22	Random Variables	advanced	skim	Each rollout $o = (o_1, \dots, o_T) \sim \pi(\cdot q)$ is a discrete-time random sequence; H_t is a deterministic function of a categorical RV.
23	Probability Distributions	advanced	read	$\pi_\theta(\cdot q, o_{<t})$ is a categorical distribution; everything in the paper is an operation on these per-position distributions.
25	Expectation	advanced	read	Shannon entropy is $\mathbb{E}[-\log p]$; GRPO advantage normalisation is per-rollout sample-mean centering.
26	Variance and Covariance	advanced	skim	Per-rollout normalised advantage $\hat{A}_i = (r_i - \mu)/\sigma$ stabilises gradient variance.
36	Shannon Entropy	advanced	drill into proof	Central object: $H_t = -\sum_v \pi_{\text{base}}(v q, o_{<t}) \log \pi_{\text{base}}(v q, o_{<t})$.
37	Cross-Entropy	advanced	read	The contrastive loss is structurally a cross-entropy between an advantage-weighted target and π_θ .
38	KL Divergence	advanced	drill into proof	Anchoring loss $\text{KL}(\pi_{\text{base}} \ \pi_\theta)$ pins π_θ to base outside decision points; the asymmetry is load-bearing.

Total: 10 nodes. Two earn the “drill into proof” tag: entropy (the paper’s organising principle) and KL (the asymmetry choice is non-obvious).

Concept pages live at <https://github.com/pleyva2004/math-foundations/blob/main/concepts/<NN>-<slug>/README.md>.

Section 3 — Paper Concepts Walk

Paper-specific objects pulled directly from 02-math-deep-dive.md.

#	Concept	What it does in the paper	Where it appears
P1	Token-level entropy H_t	Per-position predictive entropy under π_{base} . The single scalar that decides whether a position is a “decision point”.	eq. (1), §Central Object
P2	Decision-point set $D^{(i)}$	$\{t : H_t > \tau\}$. The sparse subset of positions on which RL actually changes anything; empirically 2–8% of tokens.	§Central Object
P3	GRPO advantage normalisation	$\hat{A}_i = (r_i - \mu)/\sigma$. Per-rollout variance-stabilising centering.	§Per-rollout normalised advantage
P4	Reranked vs shifted classification	A position is <i>reranked</i> if RL’s argmax differs from base’s but lies in base top-5; <i>shifted</i> if outside top-5. Headline: shifted ≈ 0.01 – 0.12% .	Table 2 / §Disagreement statistics
P5	Contrastive loss $\mathcal{L}_{\text{contrast}}$	Advantage-weighted log-likelihood, restricted to $t \in D^{(i)}$. The “do-something” half of ReasonMaxxer.	§ReasonMaxxer
P6	Base-anchoring KL $\mathcal{L}_{\text{anchor}}$	$\sum_i \sum_{t \notin D^{(i)}} \text{KL}(\pi_{\text{base}} \parallel \pi_\theta)$. The “do-no-harm” half.	§Base-anchoring KL
P7	KL-clipped policy ratio (mostly <i>discarded</i>)	Inherits group-normalisation from GRPO but discards the clipped ratio and KL-to-reference penalty.	§Note: this is the same normalisation as GRPO
P8	Rank-32 LoRA on QKVO	Full RL→base behavioural delta reproduced by a rank-32 LoRA on attention QKVO via KL distillation.	§A separate experimental thread
P9	Weight tying with input embedding	Most of the LoRA contribution concentrates in W_O ; combined with LM-head weight tying, makes the change interpretable as an embedding-space direction.	§LoRA QKVO

Section 4 — Improvements Walk

Parsed from `05-improvements.tex`. Math/Theoretical proposals are validated by PROOF (LaTeX in `proofs/`); Code/Experimental by MEASUREMENT (Python in `improvements/` with a `measure()` function whose output is checked).

Mathematical

M1 — Lipschitz bound on rank displacement. Across all four base/RL pairs Akgül report, the RL-promoted token sits at base-rank 2.14 to 2.39 — a suspiciously narrow band. Conjecture: under bounded GRPO learning rate η and per-step KL-budget β , the rank displacement at decision points is Lipschitz in (η, β) . A perturbation argument on the softmax Jacobian, combined with the operator-norm bound on the GRPO update, should yield a quantitative bound that recovers the empirical rank-2 regularity from first principles.

Validation: `PROOF` — `proofs/lipschitz-rank-displacement.tex`

M2 — Information-theoretic tightening of “top-5 suffices”. The paper observes that 99.88–99.99% of RL choices stay inside base top-5 but offers no bound. An information-theoretic argument (e.g. Fano on the reward channel) should yield k -dependent bounds of the form “with high probability under bounded KL budget, the RL choice lies in base top- k for $k \lesssim f(H_t, \beta)$.”

Validation: PROOF — proofs/info-theoretic-top-k.tex (deferred — not building this proof in v1.9)

Code (Implementation)

C1 — Hyperparameter-free quantile-based gating. Replace the absolute threshold τ (currently grid-searched per model/scale) with $D^{(i)} = \{t : H_t^{(i)} > Q_q(\{H_t\}_{t=1}^{T_i})\}$ where q is a single scale-invariant percentile. The fixed absolute $\tau = 1.4$ collapses (gates 6.2% on one prompt, 42.7% on another); the quantile gate gates exactly $1 - q$ of every prompt by construction.

Validation: MEASUREMENT — improvements/adaptive-tau.py (extended with measure()).
Expected output: {"fixed_tau_gated_pct": [6.2, 42.7], "quantile_gated_pct": [5.0, 5.0], "stability_ratio": $\approx 7x$ }.

C2 — Single-pass entropy scoring. A streaming implementation that fuses H_t computation into the forward pass (logit reuse + online running sum) is $\sim 2\times$ faster and constant-memory.

Validation: MEASUREMENT — improvements/streaming-entropy.py (NEW). Expected output: {"naive_ms": ..., "streaming_ms": ..., "speedup": ≈ 2.0 , "max_abs_diff": $< 1e-6$ }.

Experimental

E1 — Out-of-distribution stress probe. Probe ReasonMaxxer on far-OOD tasks where base success is near zero, to test whether the “no new capabilities” claim survives outside the studied regime. (*Deferred to follow-on work.*)

Validation: MEASUREMENT (deferred)

E2 — Non-verifiable task probe. The reward $r \in \{0, 1\}$ assumes a verifier; on open-ended tasks the entropy-gating story may not transfer. (*Deferred to follow-on work.*)

Validation: MEASUREMENT (deferred)

Theoretical

T1 — Active learning as the right framing. The entropy threshold τ is exactly the acquisition function of uncertainty-sampling active learning (Lewis & Gale 1994), lifted to the token level. Reframing ReasonMaxxer as “active SFT with H_t as the acquisition criterion” imports label-complexity bounds, query-efficient regret bounds, and known failure modes. (*Deferred — proof of equivalence not in v1.9.*)

Validation: PROOF — proofs/active-learning-equivalence.tex (deferred)

T2 — Low-rank LoRA delta as inference-time activation steering. The rank-32 LoRA on QKVO — concentrated in W_O — is structurally identical to representation-engineering / activation-steering interventions. A formal equivalence (LoRA delta acts at decision points as a steering vector added to the residual stream) would unify two literatures.

Validation: PROOF — proofs/lora-equiv-steering.tex

Section 5 — What to Do Next

(a) **Most personally promising proposal: M1 (Lipschitz bound on rank displacement).** The cleanest mathematician-flavoured contribution available. Empirical regularity is striking, conjecture is testable, and a one-page perturbation argument plus a softmax-Jacobian bound likely suffices. Most likely to survive peer review as a stand-alone short note because it explains an existing observation rather than proposing a new method.

(b) **Single experiment to settle the most ambiguous claim: necessity of entropy-gated correction.** The paper shows entropy-gated correction is sufficient but not necessary. Replicate ReasonMaxxer with $D^{(i)}$ replaced by (i) a random same-size subset, (ii) the complement $\{t : H_t < \tau\}$. If pass@1 collapses on both, necessity is established. One training run per ablation on a 1.5B model — a weekend’s compute.

(c) **Where to take this as a follow-on paper: bridge to mechanistic interpretability via the LoRA-as-steering equivalence (T2).** If the LoRA delta is a steering vector at decision points, then (1) extract it directly from $\pi_{\text{RL}} - \pi_{\text{base}}$ without LoRA scaffolding; (2) test whether it composes linearly across skills; (3) extract via supervised contrast on a small held-out set, fully eliminating RL. The paper writes itself: “RL for Reasoning is Activation Steering in Disguise — and Here’s the Vector.”